

LAWRENCE BERKELEY NATIONAL LABORATORY

POLYTECHNIQUE MONTREAL

Undulator benchmark (2-D)

Authors:

Gregory Giard
Christian Messe

Last modified:

September 1, 2026

The second benchmark studied for this thesis consists of a 2-D geometry of an undulator using HTS tape stacks in the grooves to generate the oscillating magnetic field. The core is made of iron, which is considered non-conducting (i.e. modelled with the ϕ formulation), with a non-linear $\mu(H)$ function. The geometry simulated is shown in Fig. 1, consisting of three pairs of poles and grooves. Each groove contains a stack of 30 fully insulated tapes, with alternating current direction between neighbouring grooves to generate the sinusoidal magnetic field in the gap. Using anti-symmetric boundary conditions on the x and y axis (see Fig. 1a), the full geometry of this model consists of two iron cores, each with six pairs of poles and grooves. This benchmark highlights the use of the ϕ formulation to model ferromagnet and the use of boundary conditions to model the symmetries. We also note that this topology features a large number of cohomology cuts (180, 30 of which are represented in Fig. 1b), automatically computed following the implementation in BELFEM. Fortunately, we note that the computation time is only influenced by the number of simplices in the mesh rather than the number of conductors, which makes the cohomology computation very fast for this 2-D benchmark (the total reduction time was 0s on this specific mesh containing approximately 112,000 elements).

The geometric parameters of the modelled undulator are presented in Tab. 1. Each tape consists of a thin-shell object containing one layer. As with the previous racetrack benchmark, the critical current density $J_c(B, \theta)$ and the $n(B, \theta)$ factor considered come from the SuperPower Advanced Pinning data in the Robinson Database [1], this time at a temperature of 15 K. The current carried by each tape is shown in Fig. 2, consisting of a 5 min ramp with an amplitude of 700 A, a 5 min plateau, a 5 min ramp down to 0 A, and a 5 min without any transport current at the end of the simulation. Considering the groove and the pole widths from Tab. 1, the wavelength of the magnetic field produced by this undulator should be 15 mm.

Fig. 3 shows the magnetic flux density distribution at $t = 10$ min (i.e. at the end of the 700 A current plateau). The simulation was conducted on a quarter of the geometry only, and this full solution representation was generated by exploiting the symmetry of the field with respect to both the $x = 0$ and $y = 0$ axes. The distribution confirms the expected operating principle of the device. The flux generated by each stack is concentrated by the two

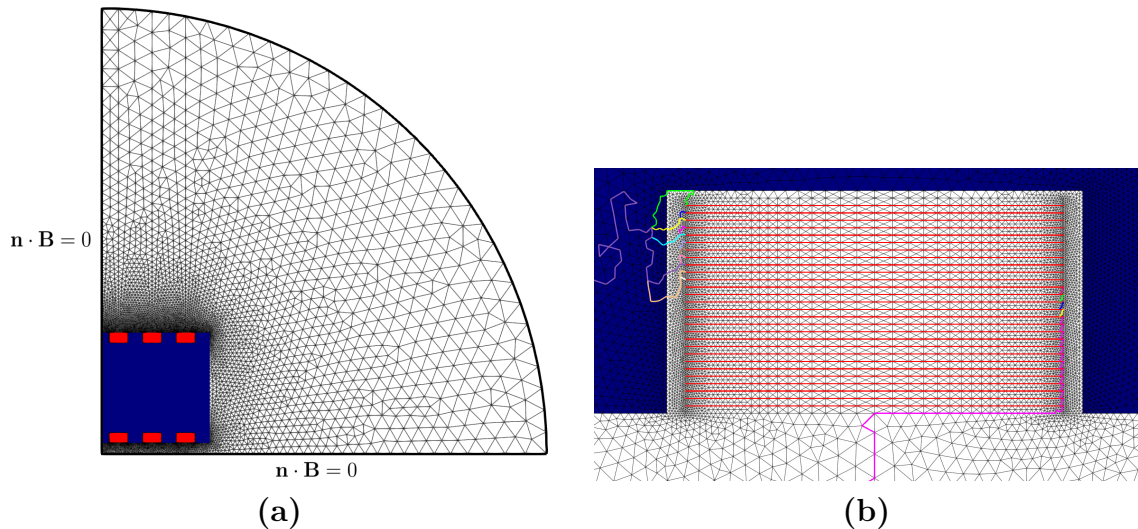


Figure 1: (a) Full geometry and mesh of the simulated undulator problem with the anti-symmetry (perfect electric conductor) boundary conditions, and (b) zoom on the bottom left groove with the cuts represented by the coloured lines. In both figures, the air is represented in white, the iron in blue and the tapes (thin-shells) in red.

Parameter	Value
Tapes width	4 mm
Tapes thickness	1 μm
Groove width	4 mm
Pole width	3 mm
Groove/pole height	2 mm
Core height	20 mm
Gap	5 mm

Table 1: Geometric parameters of the 2-D undulator benchmark.

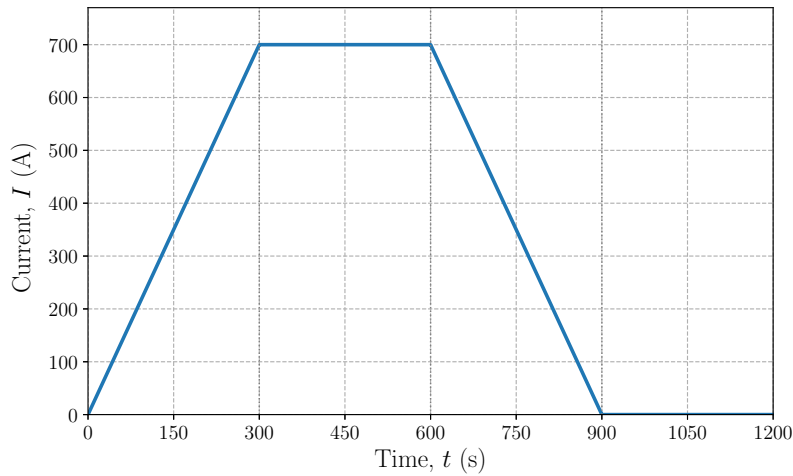


Figure 2: Current carried by each tape as a function of time. The direction of this current is alternating between neighbouring grooves in the undulator.

poles adjacent to its groove and channelled across the gap, so that neighbouring poles act as alternating north and south poles. The largest values of the flux density, up to 13 T, are found at the pole tips, where the flux lines are strongly concentrated. In these regions the iron is strongly saturated compared to the interior of the core.

The physics relevant to the undulator takes place in the middle of the gap (i.e. around the $y = 0$ axis), where a sinusoidal field is required to make the electron beam radiate. Fig. 4 presents the x component of the flux density along this horizontal axis at various times of the simulation. It is clear that the desired behaviour is indeed achieved, barring minor effects at the boundaries of the device. At the end of the current plateau, the field on the axis is sinusoidal with a peak amplitude of $B_0 \approx 3$ T and a period of $\lambda \approx 15$ mm, in agreement with the expected value from the groove and pole widths of Tab. 1.

Finally, during the last 5 min of the simulation, during which no transport current is applied, there is still a small field generated by the stacks. For Fig. 4, note that the field on the x axis does not return to 0 everywhere, but rather displays a remanent sinusoidal field of about 0 T amplitude that persists after the ramp down. Since the permeability used for the iron core is non-hysteretic, it cannot contribute to any remanence, and thus the residual field is entirely caused by the persistent currents circulating in the superconducting tapes, amplified by the magnetic circuit of within the geometry. This result is a good illustration of what

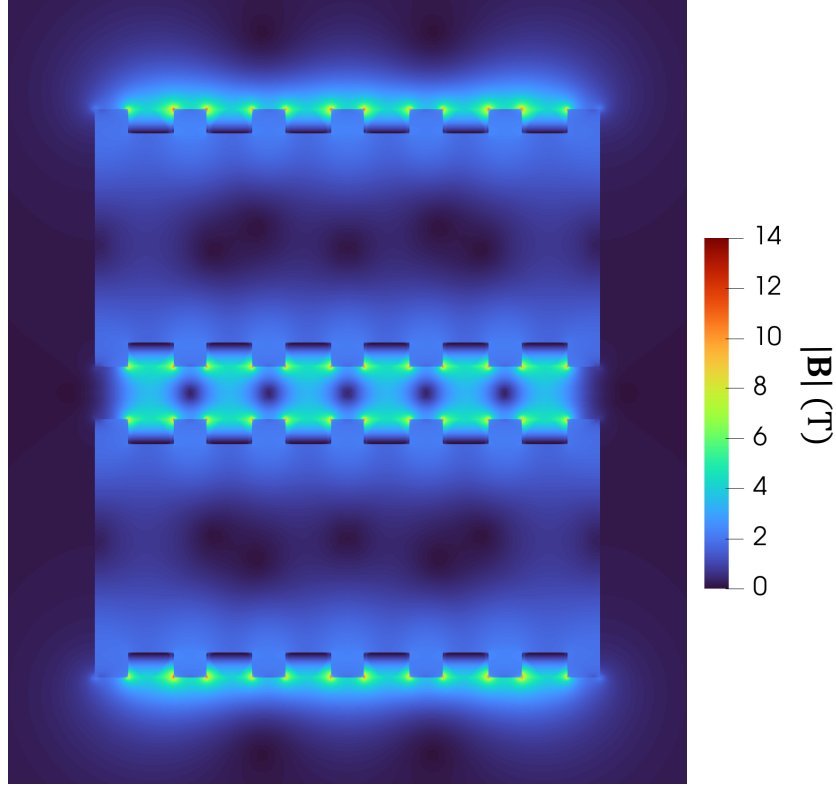


Figure 3: Magnetic flux density distribution in the undulator at $t = 10$ min.

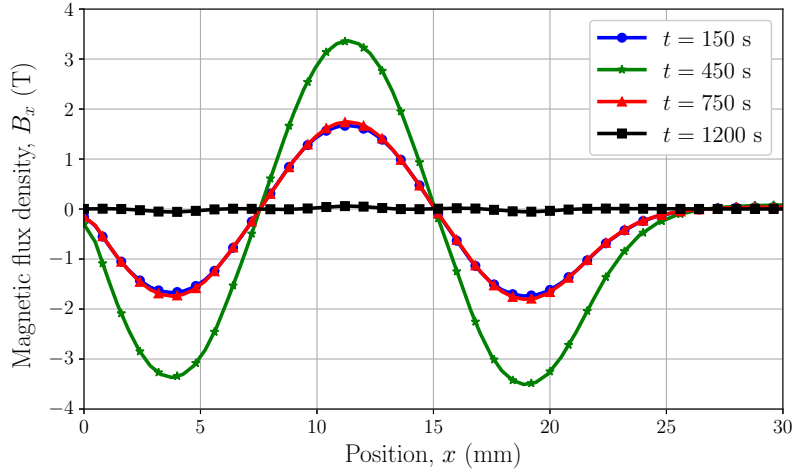


Figure 4: Distribution of the x component of the magnetic flux density along the $y = 0$ axis at various time steps.

the mixed $\mathbf{H}$ - ϕ formulation brings to this problem. The persistent currents, the non-linear response of the iron and their mutual interaction are all obtained from a single time-dependent simulation problem, without the need of an additional ad hoc magnetisation model.

1 References

- [1] S. C. Wimbush and N. M. Strickland, “A public database of high-temperature superconductor critical current data,” *IEEE Transactions on Applied Superconductivity*, vol. 27, no. 4, pp. 1–5, 2016.